

Pranav Kumar Soma

MACHINE LEARNING ENGINEER • APPLIED AI

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San Francisco Bay Area, CA • Available January 2027

EDUCATION

M.S. Computer Science, University of California San Diego <i>GPA 4.00/4.00 • Jacobs School of Engineering • completed in 1 year</i>	Jan 2026 – Dec 2026 La Jolla, CA
B.S. Computer Science, University of California San Diego <i>GPA 3.96/4.00 • Cum Laude, CSE Department Honors, Provost Honors every quarter • completed in 2 years</i>	Sep 2023 – Dec 2025 La Jolla, CA
Coursework: Data Systems for ML, Statistical NLP, Search & Optimization, Unsupervised Learning, Computer Vision, Recommender Systems, Graduate Algorithms, Parallel Computing.	

EXPERIENCE

AI Research Fellow & Agentic Systems Lead, Laboratory for Emerging Intelligence, UCSD • Fine-tuned tutoring LLMs (LoRA, SFT, RLHF) on instructor and synthetic data; deployed the AI Tutor to 3 universities across 10 courses, where a controlled study over 600+ students measured 64% fewer TA overtime hours, 87% higher engagement • Architected its two-layer eval stack: a reviewer LLM gating every student-facing response and a meta-tutor rewriting the tutor prompt from long-term signal, backed by a 5-model panel-of-judges with majority aggregation • Lead 6 engineers on a Teradata-funded agentic workbench (100+ pilot users): 60+ MCP servers over all 600 UCSD APIs plus a skill-refinement loop optimizing latency, token spend, and skill-selection accuracy	Nov 2024 – Present
Graduate Researcher, Qualcomm Institute / Calit2, HXI Lab <i>AILA, the AI Living Assistant for the LifeSaver Project</i> • Built multimodal pipelines turning sensor, audio, and video into structured health-database queries, plus a conversational interface mapping patient interaction to symptom recognition and risk assessment • Grounded diagnosis in a MongoDB vector store of UMLS codes (GraphRAG over the medical ontology); fine-tuned local Qwen models on the SD Supercomputer Center against a physician-authored golden eval set. First-author poster	Nov 2024 – Dec 2025 San Diego, CA
Honors & M.S. Thesis Researcher, UCSD CSE, Computational Redistricting <i>Advisors: Dr. Deepak Kumar, Dr. Imani Munyaka, Dr. Scott Desposato</i> • Trained per-target XGBoost regressors on per-block geospatial features with multilevel graph partitioning to generate districting plans: pooled MAE 0.042 across 9 states, held-out cross-state generalization • Implemented a Sequential Monte Carlo sampler (spanning-tree balanced cuts, N=128) and greedy DP baselines over Census 2020 data at 158K to 5.5M blocks/state ; authored efficiency-gap and VRA Section 2 metrics	Sep 2024 – Present La Jolla, CA
Software Engineer Intern, ServiceNow <i>AI Voice Agent Analytics</i> • Shipped AI Voice Agent Analytics, unifying 20+ tables of intent, agent, and assistant telemetry for 24+ enterprise instances at 10K+ voice calls/day ; cut report load 7s to under 1s over millions of records	Jun 2026 – Sep 2026 San Diego, CA
Software Engineer Intern, ServiceNow <i>AI Voice Agents • Technical Skills Award, top intern of 800+</i> • Re-architected the NLU pipeline across 8 intents and 48+ slots onto higher-capacity base models (AWS Lex V2): 45% lower agent-response latency , validated live with an enterprise customer. Return offer	Jun 2025 – Sep 2025 San Diego, CA
Founding Software Engineer, Stealth Startup <i>Autonomy & perception • part-time</i> • Building perception and autonomy : multi-sensor data-fusion pipelines, real-time object tracking, and multi-agent trajectory planning across distributed nodes, validated in simulation and hardware-in-the-loop testing	Apr 2026 – Present

SELECTED PROJECTS

- **ORCA, AI Music Production** (*Best Project, SDTC 2025: 1 of 30 from 200+; SAIRS 2025*) • hybrid neural nets for source separation, CNN audio-to-score, OpenCV vision; **98% arrangement accuracy**, SME and cross-entropy validated
- **BioVeritas** • LLM reasoning research (mechanistic, multilevel, interfield) scoring logical rigor in biology papers • **Skin Lesion Detector** • CNN vs. VLM vs. **MedGemma** fine-tuning study • **AgentSpace** • interview-derived agents over two-way **GraphRAG**

PUBLICATIONS, LEADERSHIP & HONORS

Publications: ORCA (paper, SDTC 2025; poster, SAIRS 2025), THIA (SDTC 2025), VITON (SAIRS 2025), first-author AILA poster; redistricting in prep for ACM CHI.

Co-Founder & President, Real World AI Network • **7,000+ student** federation, **50+ faculty** • **President**, CSE Society @ UCSD • **300+** members, **21+** projects.

Honors: ServiceNow Technical Skills Award • Cum Laude & CSE Department Honors • YC Startup School 2026 • National Merit Finalist • USACO Silver.

TECHNICAL SKILLS

ML & AI: PyTorch, Transformers, scikit-learn, XGBoost, OpenCV; LLM fine-tuning (LoRA, SFT, RLHF), RAG and GraphRAG, vector search, LLM agents, MCP, A2A, LLM-as-judge evals, prompt optimization, multimodal models, CNNs, LSTMs, reinforcement learning, NLP/NLU, computer vision, Sequential Monte Carlo; Qwen, Ollama, MedGemma

Languages & Data: Python, C++, C, Java, TypeScript, JavaScript, SQL, Bash; NumPy, Pandas, GeoPandas, Shapely, pymetis, Parquet/PyArrow; MongoDB (vector), PostgreSQL; multi-node GPU training, parallel computing, SD Supercomputer Center

Infrastructure: FastAPI, Flask, Node.js, Express, React; Docker, Kubernetes, Jenkins CI/CD, GCP, AWS (Lex, Lambda), Git, Linux, ROS; distributed and real-time systems, system design